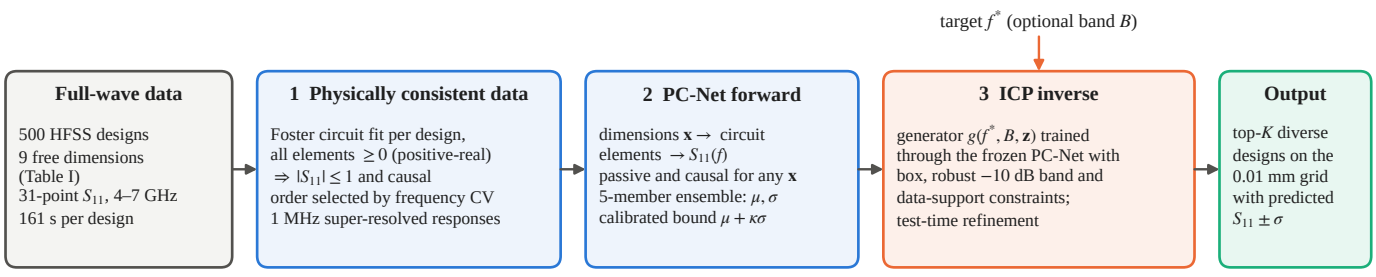
target f^* (optional band B)
Full-wave data
500 HFSS designs
9 free dimensions
(Table I)
31-point S_{11}, 4–7 GHz
161 s per design
1 Physically consistent data
Foster circuit fit per design,
all elements ≥ 0 (positive-real)
$\Rightarrow |S_{11}| \leq 1$ and causal
order selected by frequency CV
1 MHz super-resolved responses
2 PC-Net forward
dimensions $\mathbf{x} \rightarrow$ circuit
elements $\rightarrow S_{11}(f)$
passive and causal for any $\mathbf{x}$
5-member ensemble: μ, σ
calibrated bound $\mu + \kappa\sigma$
3 ICP inverse
generator $g(f^*, B, \mathbf{z})$ trained
through the frozen PC-Net with
box, robust −10 dB band and
data-support constraints;
test-time refinement
Output
top-K diverse
designs on the
0.01 mm grid
with predicted
$S_{11} \pm \sigma$